

SUSTech

Midterm for Calculus II in Spring Semester, 2019

注意：本试卷里的中文为直译（即完全按英文字面意思直接翻译），所有数学词汇的定义请参照教材（Thomas' Calculus, 13th Edition）中的定义。如果其中有些数学词汇的定义不同于中文书籍（比方说同济大学的高等数学教材）里的定义，以教材（Thomas' Calculus, 13th Edition）中的定义为准。

1. (15 pts) Determine whether the following statements are **true** or **false**? No justification is necessary.

- (1) If $a_n < 0$ for $n > 1000$, and $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ may diverge.
- (2) Suppose that the power series $\sum_{n=1}^{\infty} a_n(x-1)^n$ converges at $x = 0$, and diverges at $x = 2$, then the interval of convergence of this series is $[0, 2)$.
- (3) If $f(x, y)$ has partial derivatives $f_x(x, y)$ and $f_y(x, y)$ at (x_0, y_0) , then

$$\lim_{x \rightarrow x_0} f(x, y_0) = \lim_{y \rightarrow y_0} f(x_0, y) = f(x_0, y_0).$$

- (4) If a vector function $\mathbf{r}(t)$ is always perpendicular to its derivative $\frac{d\mathbf{r}}{dt}$, then $|\mathbf{r}(t)|$ must be constant.
- (5) The curvature of a unit circle is greater than the curvature of the parabola $y = x^2$ at the origin.

一、（15分）判断题：

- (1) 若当 $n > 1000$ 时 $a_n < 0$ ，并且已知 $\sum_{n=1}^{\infty} a_n$ 收敛，则 $\sum_{n=1}^{\infty} a_n^2$ 有可能发散。
- (2) 若幂级数 $\sum_{n=1}^{\infty} a_n(x-1)^n$ 在 $x = 0$ 处收敛，而在 $x = 2$ 处发散，那么该级数的收敛区间是 $[0, 2)$ 。
- (3) 若函数 $f(x, y)$ 在 (x_0, y_0) 处存在偏导数 $f_x(x, y)$ 和 $f_y(x, y)$ ，那么

$$\lim_{x \rightarrow x_0} f(x, y_0) = \lim_{y \rightarrow y_0} f(x_0, y) = f(x_0, y_0).$$

- (4) 若一个向量函数 $\mathbf{r}(t)$ 总是垂直于其导数 $\frac{d\mathbf{r}}{dt}$ ，则 $|\mathbf{r}(t)|$ 一定是常数。
- (5) 在原点处，单位圆的曲率大于抛物线 $y = x^2$ 的曲率。

2. (9pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

- (1) Let $\mathbf{a}$ and $\mathbf{b}$ be two nonzero orthogonal vectors, which of the following must be true?
 (A) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$; (B) $|\mathbf{a} - \mathbf{b}| = |\mathbf{a}| - |\mathbf{b}|$;
 (C) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$; (D) $\mathbf{a} + \mathbf{b} = \mathbf{a} - \mathbf{b}$.
- (2) The equations of two lines are $l_1: x = t, y = 2t, z = -t$, and $l_2: x = 1 - 2t, y = t, z = -1 + t$. Then l_1 and l_2 are
 (A) parallel; (B) orthogonal; (C) intersect with each other; (D) skew.
- (3) Suppose $0 \leq a_n < \frac{1}{n}$, ($n = 1, 2, \dots$), then which of the following series must converge?
 (A) $\sum_{n=1}^{\infty} a_n$; (B) $\sum_{n=1}^{\infty} (-1)^n a_n$; (C) $\sum_{n=1}^{\infty} \sqrt{a_n}$; (D) $\sum_{n=1}^{\infty} (-1)^n a_n^2$.

二、(9分) 单项选择题:

- (1) 若向量 $\mathbf{a}$ 和 $\mathbf{b}$ 都是非零正交向量, 下面哪一个结论是正确的?
 (A) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$; (B) $|\mathbf{a} - \mathbf{b}| = |\mathbf{a}| - |\mathbf{b}|$;
 (C) $|\mathbf{a} + \mathbf{b}| = |\mathbf{a} - \mathbf{b}|$; (D) $\mathbf{a} + \mathbf{b} = \mathbf{a} - \mathbf{b}$.
- (2) 设两条直线的参数方程为 $l_1: x = t, y = 2t, z = -t$, 以及 $l_2: x = 1 - 2t, y = t, z = -1 + t$. 那么 l_1 与 l_2 的关系为
 (A) 平行; (B) 正交; (C) 相交; (D) 异面.
- (3) 若 $0 \leq a_n < \frac{1}{n}$, ($n = 1, 2, \dots$), 那么下面哪一个级数一定收敛?
 (A) $\sum_{n=1}^{\infty} a_n$; (B) $\sum_{n=1}^{\infty} (-1)^n a_n$; (C) $\sum_{n=1}^{\infty} \sqrt{a_n}$; (D) $\sum_{n=1}^{\infty} (-1)^n a_n^2$.

3. (9 pts) Does the following series absolutely converge, conditionally converge, or diverge? Give reasons for your answer.

- (1) $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$.
- (2) $\sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}}$.
- (3) $\sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}$.

三、(9分) 下列级数是否绝对收敛、条件收敛或者发散? 给出你的理由.

- (1) $\sum_{n=1}^{\infty} (-1)^n \left(1 - \frac{1}{n}\right)^{n^2}$.
- (2) $\sum_{n=1}^{\infty} (-1)^n e^{-\frac{1}{n}}$.

$$(3) \sum_{n=2019}^{\infty} (-1)^n \frac{1}{\sqrt{n^2 - 2019n + 1}}.$$

4. (10 pts)

(1) Find the radius and interval of convergence of the series

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n x^n}{\sqrt{n^2 + n + 1}}.$$

(2) For what values of x does the series converge absolutely, or conditionally?

四、 (10分)

(1) 求级数

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} 2^n x^n}{\sqrt{n^2 + n + 1}}$$

的收敛半径和收敛区间.

(2) x 取哪些值时级数绝对收敛, 取哪些值时级数条件收敛?

5. (10 pts) Let $x = \cos^3 t$, $y = \sin^3 t$, where $0 \leq t \leq \frac{\pi}{2}$, be a parametrization of a curve.

(1) Find the length of the curve.

(2) Find the area of the surface generated by revolving the curve about the x -axis.

五、 (10分) 设 $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq \frac{\pi}{2}$, 为一参数化曲线.

(1) 求曲线的弧长.

(2) 求曲线绕 x 轴旋转所得到的曲面的面积.

6. (10 pts) Find the equation of the plane through the points $(2, -1, -1)$ and $(1, 0, -1)$ perpendicular to the plane $2x + 3y - 5z + 6 = 0$.

六、 (10分) 如果一个平面通过两个点 $(2, -1, -1)$ 和 $(1, 0, -1)$, 并且与平面 $2x + 3y - 5z + 6 = 0$ 垂直, 请写出平面方程.

7. (10 pts) A particle is located at the point $(1, 0, -2)$. Its initial speed is $|\mathbf{v}(0)| = 3$ at time $t = 0$, and the direction of its initial velocity is toward the point $(2, -1, 3)$. The particle moves with constant acceleration $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. Find its position vector $\mathbf{r}(t)$ at time t .

七、 (10分) 一个物体在 $t = 0$ 时初始坐标为 $(1, 0, -2)$. 它在 $t = 0$ 时的初始速率为 $|\mathbf{v}(0)| = 3$, 而且其初始速度的方向是指向点 $(2, -1, 3)$. 这个物体运动的加速度为常数向量, $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$. 请写出其位移向量 $\mathbf{r}(t)$ 关于时间 t 的参数表示.

8. (10 pts) Is the following function, $f(x, y)$ continuous at $(0, 0)$? Give reasons for your answer.

$$f(x, y) = \begin{cases} \frac{\sin(x^3+y^3)}{x^2+y^2}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$$

八、 (10分) 请问如下函数 $f(x, y)$ 在 $(0, 0)$ 处是否连续, 给出你的理由.

$$f(x, y) = \begin{cases} \frac{\sin(x^3+y^3)}{x^2+y^2}, & (x, y) \neq (0, 0); \\ 0, & (x, y) = (0, 0). \end{cases}$$

9. (9 pts) Let $f(u)$ be differentiable, $z = f(e^x y)$ ($y \neq 0$), and $\frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 1$. If $f(1) = 0$, find $f(u)$.

九、 (9分) 设函数 $f(u)$ 可导, 定义 $z = f(e^x y)$ ($y \neq 0$), 而且有 $\frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 1$. 已知 $f(1) = 0$, 请写出 $f(u)$ 的表达式.

10. (8 pts) Find the Taylor series for $f(x) = \ln(x + \sqrt{x^2 + 1})$ at $x = 0$.

十、 (8分) 请写出函数 $f(x) = \ln(x + \sqrt{x^2 + 1})$ 在 $x = 0$ 处的Taylor级数.